Geoforum 150 (2024) 103977

Contents lists available at ScienceDirect

Geoforum

journal homepage: www.elsevier.com/locate/geoforum

Water trading markets: Facilitating financial flows through the hydro-social cycle?

Madelaine Moore

Bielefeld University, Universitätsstr 25, Bielefeld, 33615 Germany

ARTICLE INFO

Keywords:
Water trading market
Hydro-social cycle
Financialization
Water governance
Futures market
Water crisis

ABSTRACT

The world is facing a global water crisis, a crisis that is increasingly framed as a crisis of water governance. As water activists have successfully problematised neoliberal water governance, the question about what is going to come next remains. While current governance proposals appear somewhat similar to what came before, one key shift is the increasing articulation of water governance to circuits of global finance, seen through a growth in financial investors in the water sector but also the way that water and water services and infrastructures are being rendered legible for inclusion into financial markets – the first water futures market being one example. This article explores the water futures market in dialogue with three long-standing water trading markets (Australia, USA and Chile) in order to untangle the processes involved in, and barriers to, the financialization of water resources. The argument put forward is that although material barriers to the expansion of a water futures market exist, this does not mean that water resources remain immune to these processes. Instead, logics of crisis, risk management, and questions of financing appear to characterise this shifting terrain of global water governance across the hydro-social cycle. Moreover, when explored as a hydro-social cycle the financialization of these nodes are not singular "one-offs" but may instead feed into, facilitate, or conversely hinder further financialization across the hydro-social cycle.

1. Introduction

At the end of 2020 the world's first water futures market was announced in California – the Nasdaq Veles California Water Index. It claimed to help water users manage water (in)security through the purchase of water futures (CME Group, 2021). In response Willem Buiter, former chief economist of financial firm Citigroup stated that although in 2011 he had expected (Buiter, 2021):

> 'a globally integrated market for fresh water within 25 to 30 years. Once the spot markets for water are integrated, futures markets and other derivative water-based financial instruments — puts, calls, swaps — both exchange-traded and OTC will follow. There will be different grades and types of fresh water, just the way we have light sweet and heavy sour crude oil today… Water as an asset class will, in my view, become eventually the single most important physical-commodity based asset class, dwarfing oil, copper, agricultural commodities and precious metals.'

The current futures market was 'somewhat premature' as the 'spot market underlying the futures market is too segmented; it does not represent a single, homogeneous commodity or asset. Today's spot markets for water and water rights thus are too illiquid and non-transparent to support an economically and socially useful futures market.' Highlighting, if indirectly, the current material barriers to the financialization of water.

However, the problems that the futures market seeks to address (increased risk, scarcity, competition…) are reflective of a growing global water crisis and a perceived crisis of global water governance (UNESCO & UN Water, 2021; UNESCO & UN Water, 2023). For example, recent estimates suggest that over 2 billion people still lack access to potable water (UNESCO & UN Water, 2023, p. 1), and in 2018 14 of the world's 20 mega cities were deemed water stressed (Leahy, 2018). Despite policy proposals such as privatisation and the commercialisation of nominally public water infrastructures and services being central to what can be called neoliberal water governance (Moore, 2023; Muehlebach, 2023), current proposals to address our current crisis of governance appear similar; Public-Private Partnerships (PPPs) and blended finance at one node[1] of the hydro-social cycle, and Water

E-mail address: madelaine.moore@uni-bielefeld.de.

[1] The term node refers here to different forms moments across the hydro-social cycle. For example, water services, pipelines, irrigation, aquifers and so on. They are all parts of the whole (hydro-social cycle) and entangled with one another but often approached as separate entites.

https://doi.org/10.1016/j.geoforum.2024.103977
Received 3 February 2023; Received in revised form 13 February 2024; Accepted 20 February 2024
Available online 21 March 2024
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Trading Markets (WTMs), water banks or now futures markets at another. The focus is on how scarcity can be managed through economic logics and tools, which are presented as efficient, equitable, and to some degree neutral (CME Group, 2021, p. 3). However, one key shift from neoliberal water governance to today is the increasing articulation of water governance to circuits of global finance, seen through a growth in financial investors in the water sector but also the way that water and water services and infrastructures are being rendered legible for inclusion into financial markets – the futures market being one example. The argument to be developed in what is to follow, is that although water trading has existed for centuries (see Trawick et al., 2014), in recent decades some of the most developed WTMs have been re-regulated for profit-seeking and subsequently integrated into circuits of global capital, suggesting that the characteristics of the futures market, alongside the barriers that Buiter outlines, are not as novel as initially thought. Put simply, logics of crisis, risk management, and questions of financing appear to characterise this shifting terrain of global water governance across the hydro-social cycle.

Understanding the hydrological cycle as a hydro-*social* cycle helps account for the way water is shaped by distribution networks but also flows of global capital (Furlong & Kooy, 2017, p. 892; Swyngedouw, 2009). The concept has recently travelled beyond its political ecology and human geography origins; in October 2022, the American Geological Society updated their hydrological cycle graphics to include, for the first time, human activities such as infrastructures, irrigation, urban centres and production (USGS, 2022). As Ballestero suggests (Ballestero, 2019, p. 407), there is a growing recognition that water's materiality is both social and natural, each co-constituting one another. Yet thinking about water as a hydro-social *cycle* also allows us to see how water flows through water services, aquifers, rivers, and pipelines, highlighting the relations between these nodes and how processes such as financialization may flow between them. And while a hydro-social *regime* is a relatively stable constellation of these relations (Goldman & Narayan, 2019, p. 96), the shifts in water governance outlined above suggest this regime may be in flux. Yet, as Goldman and Nyman argue (Goldman & Narayan, 2019, p. 96), there may not be a clear break from one regime to the next but instead the new regime is 'nested and overlapping' with what came before.

Exploring the hydro-social cycle as a whole it will become clear that as water flows through catchment zones, to irrigation, water services or infrastructures it is being increasingly embedded into global circuits of financial capital. Turning to WTMs, financial actors are increasingly buying entitlements and WTMs are being read through financial logics (such as speculation and risk management). This is, as I will show, articulated through logics of crisis, environmental protection, and efficient measurement and monitoring. As such, although material barriers to the expansion of a water futures market exist, as Buiter points out (Buiter, 2021), this does not mean that water resources remain immune to these processes. Moreover, when explored as a hydro-social cycle the financialization of these nodes – understood as the different forms of water (services, pipelines, rivers or aquifers) – are not singular "one-offs" but may instead be embedded within one another, for example, when financial actors enter WTMs pushing up prices this may encourage more financial investors and investments in water services or infrastructures further along the hydro-social cycle. And in thinking through what is being replaced but also stabilising in the next hydro-social regime, it will become clear that the tendencies within WTMs and the futures market reflect existing trends across the hydro-social cycle, namely, an attempt by global finance to abstract water from its hydro-social particularities.

To develop this argument, I will first explore the current literature on financialization, in particular focussing on the debates on water financialization. I will then use these debates to situate the futures market in dialogue with the three most developed WTMs globally, namely, the Murray Darling Basin in Australia, the West Coast of the USA, and Chile. By bringing these cases together through a relational comparison (see Bruff, 2021; McMichael, 1990) I am then able to draw out common tendencies but also the potential contradictions and barriers that emerge through their orientation towards circuits of global finance. In exploring these different WTMs and the futures market in a relational comparison this paper seeks to build upon existing comparative studies of WTMs (Easter et al., 1998; Garrick, 2015; Grafton et al., 2011; Wheeler, 2021), while allowing me to make some more broader claims about the shifting terrain of global water governance.

2. What is Financialization?

Financialization has been subject to various interpretations, encompassing logics of economic efficiency, risk management, short-term profit maximization, and speculation, alongside the increasing influence of financial interests, actors, markets, and institutions in the domestic and international economy. And although water has been, as Bakker describes (Bakker, 2004), an uncooperative commodity, it is not immune to processes of enclosure and financialization. The following aims to explore the concept of financialization and its application to nature and water specifically.

2.1. Financialization

As Mader, Mertens and Zwan assert (2020) the concept of financialization is somewhat nebulous. One of the most widely cited definitions comes from Epstein who states (Epstein, 2005, p. 3), that financialization highlights the way that financial interests, actors, markets and institutions play an increasing role in the domestic and international economy. As Fine and Christophers propose (2020), financialization can be further specified as either a form of governance dominated by logics such as economic efficiency, risk management, short-term profit maximisation, and, critically, speculation (i.e. a qualitative shift), or as a way to distinguish the quantity of profits accumulated by finance capital (i.e. a quantitative shift) (see for example, Krippner, 2005, pp. 174–175). Research can be divided into three broad foci: macro changes in capitalist accumulation and/or the state-market dichotomy, meso level changes in ownership and control of financial markets, or the micro level experiences and impact of financialization on everyday life (Mader et al., 2020). It is generally agreed that financialization goes beyond the economic, shaping our everyday life, and that it has resulted in increasing inequality, precarity, a lack of accountability, and democratic deficits (Cleaver, 2017; Langley, 2020).

The "newness" of financialization is also debated. According to Fine and Christophers (2020) financialization should be approached as an intensification of existing capitalist processes rather than a qualitative break with previous forms of value accumulation. As Banaji shows (Banaji, 2020), finance capital has historically lubricated the circulation of value, yet what is "new" is the intensification of these tendencies and actors within the global political economy. Logics of risk management, speculation, and rent have gained prominence. As Langley has argued, and drawing on Lisa Adkins work (Langley, 2020, p. 71), risk management is not about the diminishing of risk, but rather its transformation into a speculative asset that recalibrates our relation to the past, present and future. This consolidates the role of new actors such as ratings agencies and financial services industries that construct further instruments to determine risks and yields and "produce" new financial products based upon these – the water futures market being one example. Spaces, natures, and social relations are re-engineered in order to be made legible for financial investors and facilitate inclusion into circuits of global finance. As Chiapello suggests (Chiapello, 2020, pp. 84–85) this is a multi-step process involving first the creation of a *problem* that can be solved through investment (for example, the lack of water access as an issue of financing), second, where these problems are made *tangible* and can be included in systems of account (for example, the creation of water licences and quantified water resources), and third, *financial (re)structuring* where operations are re-organised so that they are open to profit seeking at each point in the accumulation process (for

example, new water infrastructures or the futures market). Importantly, understanding financialization as a process focusses on what is being *done* rather than the actors involved or degree of financialization that results (Chiapello et al., 2020, p. 82).

2.2. The financialization of nature

Nature has often been approached as a new frontier[2] of financialization where socio-nature relations are brought into the sphere of finance through new products such as eco-system services or carbon trading (Bryant, 2018). For example, eco-system services turn nature into a source of rent, facilitated through a mix of public, private and financial actors, which makes it increasingly harder to maintain a clear public–private categorisation. Terms like Natural Capital and eco-system services seek to mobilise accounting tools to securitise revenue streams from payments for eco-system services, often through the creation of green bonds that 'allow borrowing from investors with the debt secured on the future economic and environmental benefits predicted to flow from these investments.'(Bresnihan et al., 2020, p. 135). In these examples, it is the service (aquifer recharge, or Co2 sequestration through forests) rather than the trees or swamp itself that is being capitalized (Ballestero, 2023, pp. 7–9). These developments, as Bresnihan argues (Bresnihan et al., 2020, p. 124), have occurred through narratives of nature scarcity and a techno-managerial response to ecological (and arguably economic) crisis. These fictitious commodities are produced through an assemblage of science, technology, and state and non-state intermediaries, which has enabled nature to be abstracted from its materiality and place and thus be more easily integrated into circuits of global finance (Bresnihan et al., 2020, p. 136).

2.3. Financialization of the hydro-social cycle

Turning more concretely to the financialization of the hydro-social cycle, there is a growing literature on the financialization of water services and infrastructure (see Bayliss, 2014; Loftus, March and Purcell, 2018; Pryke and Allen, 2019). This builds on extensive research on the neoliberalization of water management that can be characterised by the increasing commercialisation of nominally public water services, metering, full cost recovery, and the spread of water markets (Acevedo Guerrero et al., 2015, p.173).[3] Research on water service and infrastructure financialization tends to focus on the re-making of infrastructures as an asset class (see Bresnihan, 2016; Loftus et al., 2018; Pryke & Allen, 2019). Although Furlong (Furlong, 2020, p. 574) has also pointed attention to the ways in which government enterprises have themselves become financial investors and are dependent upon financial incomes, which further problematises the public–private dichotomy that dominated previous debates on the privatization of water services and infrastructures (see Bieler, 2014; Muehlebach, 2023).

Exploring how water infrastructures are re-made into an asset class, Pryke and Allen show (Pryke & Allen, 2019) how infrastructures such as desalination plants are being orientated towards the interests of private financial capital. As others have argued (Bresnihan, 2016, p. 116; Loftus et al., 2019; Pryke & Allen, 2019), water infrastructures are attractive to institutional investors and pension funds because they offer a captive income stream – they profit from a distance through forms of financial rent. Or as Muehlebach states (Muehlebach, 2023 p.3) the water (or vital) frontier is one of revaluation where localised qualities are turned into abstract quantities through global regimes of value making. To do this these infrastructures and services must be turned into a legible financial asset, yet, and further blurring the lines of public–private, the infrastructure or service does not need to be privatised before this occurs[4]: A key revenue stream is the water bill which can be securitised, public actors can enter financial markets to attract financing, and new infrastructures can be being financially re-engineered to allow for increasing service and management fees to be extracted (for a detailed example of this process see Pryke and Allen (2019, p. 1327). As Pryke and Allen demonstrate (2019, p. 1329), these financialised infrastructures (in their case a desalination plant) can be understood as not only feats of civil but rather financial engineering due to their capacity to find, construct, and appropriate value.

In terms of the hydro-social cycle, this financial re-engineering requires us to move beyond the typical infrastructures and services that water flows through to also include the 'computer screens, labs, government institutions, financial spreadsheets, and congressional bodies…' that are necessary for this re-engineering to occur (Ballestero, 2019, p. 406). As a result, financialization processes have enabled illiquid forms of assets such as water infrastructures to be transformed into liquid assets with rent flowing from monopoly ownership of a basic need (Loftus et al., 2019, p. 1). Water-based financial products such as water targeted investment funds, structured water products, and exchange-traded funds that focus on water services and infrastructures have subsequently emerged (Bayliss, 2014, p. 292-5).

In the above examples, it is not the resource itself but rather the relation between the resource and service or infrastructure that results in a price and subsequent market orientation. Yet starting from another node in the hydro-social cycle – namely water resources such as rivers, aquifers, or groundwater – there are similar tendencies at work. As Schmidt and Matthews show (Schmidt & Matthews, 2018, p. 151), there has been a shift in water governance from a focus on water scarcity and Integrated Water Resource Management (IWRM) to the water-energy-food-climate nexus,[5] which is dominated by securitisation and techniques of risk management, prediction, and subsequent speculation. Where IWRM was water focused and its scale was the watershed, the nexus allows for water governance to be firmly articulated to the green economy and concurrent financial and ecological crises (Schmidt & Matthews, 2018, p. 153). Reflecting these claims, Ballestero's research on Water Trusts further demonstrates (Ballestero, 2023, p. 2) the specific mechanisms by which water resources are being made legible for financial actors, specifically as an environmental service or potential environmental impact investment. This articulation of water resource governance to questions of financing also dominated the 2023 UN Water Conference, the first UN conference dedicated to water in over fifty years (United Nations, 2023).

In summary, financialization is a multifaceted and ongoing process that reshapes economic, social, and ecological relations. Turning to the hydro-social cycle, financialization transforms water services, infrastructures, and resources into potential financial assets, attracting financial investors and leading to increasing securitization and speculation across the hydro-social cycle. Turning concretely to WTMs, the following section will explore the mechanisms through which water is being turned into an asset class through WTMs, but also the barriers that stand in the way of this process.

3. Water Trading Markets

In contrast to water services and infrastructures, WTMs represent the

[2] Ballestero (Ballestero, 2023) challenges the idea of a frontier, arguing that this implies a uni-directionality and instead nature financialization fluctuates unevenly and sometimes in contradictory ways.

[3] However, as Acevedo Guerrero et al demonstrate, and affirming the above argument that hydro-social regimes tend to overlap with one another (Goldman & Narayan, 2019, p. 96), these developments are not unique to the neoliberal period of water governance (Acevedo Guerrero et al., 2015).

[4] Bresnihan (Bresnihan, 2016) illustrates this process well in his analysis of Irish Water as a nominally public water services company orientated towards global financial markets.

[5] In the 2023 UN Water Development Report this was updated to the food-water-energy-ecosystem nexus (UNESCO & UN Water, 2023).

exchange and valuation of water itself. What is traded is the right to use a given amount of water, often called an allocation, licence, or contract for a specific amount of time (Varghese, 2013, p. 3). The creation of a licence is a private property right claim to a specific body of water. Although historically connected to land and riparian rights, there is a global trend towards the unbundling of water and land property rights, in order for water entitlements to become their own commodity (Grafton et al., 2011).[6] Although each WTM is a specific governance response to often localised problems, they are based on the same fundamentals; water and land are defined as separate commodities, a market is created to allow water users to trade a specific amount of water in a designated area, and water prices are determined through the market. To do this, water – what is to be traded – must be standardised, and accurately measured and monitored by institutions which are often state-led. It is hoped that these regimes will both decrease water use through encouraging water efficiencies (see WaterFind, 2018) but also add needed flexibility to existing water governance regimes (Garrick, 2015). Following Chiapello's processual understanding of financialization, when the global water crisis is problematized as an issue of governance, WTMs are framed as a governance solution through which prices can be more efficiently determined, climate risk and water insecurity managed, and tensions between environmental and social-economic objectives managed. As will be shown in each of the cases explored below this has resulted in an increasing orientation towards circuits of global finance.[7]

3.1. The water futures market

On December 7, 2020, a tradable water price futures index was launched on the Chicago Stock Exchange. This index is based on the Nasdaq Veles California Water Index (NQH2O), with each contract representing 10 acre-feet of water.[8] The index sets a weekly benchmark spot price for water rights in California, based on the average transaction prices in California's five largest and active WTMs. Nasdaq developed the NQH2O Index in partnership with Veles Water Limited, a firm specializing in the development of financial products for water markets. The data utilized in the calculation of the index is provided by West-Water Research, an economic consulting firm focused on pricing, valuation, and transaction services for water rights (CME Group, 2022). The arguments used to justify this launch were threefold: it would be a tool for managing climate risk, would provide agricultural, commercial, and municipal water users with greater price transparency, and like other futures markets, water users could use it to manage future price risks and price discovery (Suppan, 2020), allowing agricultural users to 'plan ahead' (Saefong, 2020). As Deane Dray, a managing director at RBC Capital Markets states (quoted in Saefong, 2020), 'If this new futures contract shows promise, it could spawn more innovation in the futures market related to water.' The futures market is open to all types of investors (not only water users) meaning that water is being treated like any other commodity that is currently traded on futures markets. With the futures market, the "problem" of water scarcity has evolved from one of competing users and allocation, to risk management across time and increasingly space.

3.2. West coast USA

As stated above, the futures market builds on existing Californian WTMs or put differently, Californian WTMs were a necessary precondition for the futures market to exist. WTMs expanded across western US states (specifically the Colorado and Columbia basins) during a multiyear drought in the late 1980s and early 1990s. Water trading was deemed an equitable way for those with existing water licences to profit from water savings while also offering flexibility within the historical first-in-time first-in-line system that prioritises historical licences over newer market entrants (Garrick, 2015, p. 93-4). Water rights primarily cover surface rather than ground water. Reflecting Buiter's argument that water trading is currently 'segmented' in part because of the material barriers of water's hydrology, trades are primarily limited within water districts. Auctions are used to link buyers and sellers and determine prices, and what is traded is the right to use water rather than the water itself, reflecting the Australian rather than Chilean model (more detail below). To try to counter potential speculation, Californian water licences expire after five years (2021 California Water Code Chapter 1.5., 2021), and there is a distinction made between wet rights and paper rights; water-rights-holders must demonstrate that the water they propose to lease or sell is water they would have otherwise used i.e., is a real water saving (Lachman et al., 2016).

While trading initially focussed on agricultural users it has since opened up to other users such as municipalities and extractive industries (Barlow, 2013, pp. 79–80; Lachman et al., 2016, pp. 144–146). Cities and urban water users are increasingly entering the market and buying up water rights from agricultural users or buying land in order to gain the attached water licenses – highlighting the intertwined nature of water and land grabbing (Stockton, 2016). Quoted in the New York Times (Howe, 2021), 'Matthew Diserio, the president and co-founder of the hedge fund Water Asset Management, has called the U.S. water business "the biggest emerging market on earth" and "a trillion-dollar market opportunity."' Despite this, there are concerns around the scalability and institutional requirements of US markets as water rights are hard to standardise, and because trades are largely limited to water users (wet rights), buyers and sellers are limited to specific catchment zones (Garrick, 2015, p. 100; Lachman et al., 2016).

3.3. Australia

Like other water scarce states, water policy in Australia has been highly contentious. In Australia WTMs are one tool in a more comprehensive water management plan,[9] which initially focussed on an over-allocation of water licences (and subsequent over-use) of the Murray-Darling Basin (MDB) (National Water Commission, 2011, p. 23).[10] By the 1960s it was clear that the MDB was over-allocated and restrictions on new water entitlements began to be issued, which led to informal MDB WTMs from the late 1960s onwards (S. Wheeler, Loch, et al., 2014, pp. 32–33). Yet it was not until the 1990s that a formalised cap-and-trade system developed (S. Wheeler, Bjornlund, et al., 2014), where the number of available water licences was capped and the government, acting on behalf of the environment, would reduce the number of water licences and water allocations over time. Trading was proposed as the most efficient mechanism through which scarce water could be allocated to high value users that optimised 'economic, social, and environmental outcomes,' using market-based instruments (O'Brien, 2008, p. 521).

In Australia, a water entitlement gives the bearer permission to

[6] Water rights become commodities in two senses of the term, they are produced through commodified labour and can be exchanged on the market in their own right (see Moore, 2022 for a discussion on water commodification).

[7] According to environmental NGO, Nature Conservancy, 37 countries have water rights regimes that with the right enabling conditions could evolve into WTMs (Richter, 2016, p. 7), which could result in annual water trades worth USD$13.4 billion and water assets (primarily water licences) of USD$331 billion (Richter, 2016, p. 16).

[8] An acre-foot of water is the standard measurement of a volume of water in the US and means the amount of water required to cover an acre of land with one foot of water.

[9] Other water governance policies include subsidies for water efficient infrastructure, and government buy-backs of water licences (Grafton, 2019, p. 117).

[10] The Murray Darling Basin is Australia's largest river basin that feeds into the Great Artesian Basin and cuts across multiple states.

extract a designated portion of the general pool of water available each year, rather than a fixed volume. As such, the volume of water differs each year according to water availability and is also dependent on whether the entitlement is classed as high or low security (Grafton, 2019, p. 119; National Water Commission, 2011, pp. 15–16). There are two key types of trades; water access entitlements which are a permanent trade and can be rated as high, general or low security, and water allocations which are the water assigned to an entitlement in a given year (Horne and Grafton, 2019 p. 167). Trading occurs through private water brokers and water exchanges such as Waterexchange, which advertise the current market-driven price for different water entitlements and allocations depending on the catchment and security rating.

During the 1990s trading was limited to intra-district trades, yet reforms throughout the early 2000s allowed for inter-district trades and non-productive users – speculators – to enter the market (Horne and Grafton, 2019 p. 169) – a key step towards overcoming the material barriers to abstracting water from its hydro-social conditions. Although the market was historically developed with irrigators as the key actors (Horne and Grafton, 2019 p.172), the market has evolved to consider the environment and Cultural Flows for the Murray Lower Darling Rivers Indigenous Nations (MLDRIN) and Northern Basin Aboriginal Nations (NBAN) (Australian Cultural Heritage Management, 2014). Buyers and sellers now include the government on behalf of the environment, extractive industries, environmental NGOs, First Nations and Traditional Owners, and investors (Department of Agriculture, 2016; O'Donnell et al., 2021). In times of drought, some urban and regional water authorities have purchased allocations and entitlements (S. Wheeler, Loch, et al., 2014, p. 36), however, unlike the US, there is less inter-sectoral trade between agriculture and urban water supplies (Grafton et al., 2011, p. 13). Today the southern MDB is the most developed WTM in the world and surface water is almost completely covered by tradeable water rights (National Water Commission, 2014, p. 2). Fledgling markets have begun for water delivery rights, forward allocation markets, water entitlement leases, flood plain harvesting, and carryover capacity (Horne and Grafton, 2019 p.176).

3.4. Chile

The Chilean WTM most closely reflects that of a private good rather than a cap-and-trade system designed to manage environmental scarcity (i.e., Australia). Under Pinochet's reforms of the 1981 Water Code, promoted by the IMF and World Bank, water management included strong private water property rights and market-based mechanisms, teamed with weak government regulation and oversight (Bauer, 2008, p. 3). Institutional frameworks were designed to fit and support WTMs, rather than WTMs being one of many water governance tools (Bauer, 2005, p. 148; Budds, 2004). Irrigation and agriculture were the key focus of the reforms with little attention paid to non-agricultural users or environmental protection.

The Water Code declares that water is an inalienable public property to which the national government can grant private parties exclusive rights through the Dirección General de Aguas (DGA), a government body that administers water rights but has no authority to address conflict between users (Budds, 2009, p. 421). If there is water to be allocated, water rights are given free of charge and if there are not enough new rights for all users then water rights are auctioned off. The DGA has no knowledge of how the water will be used, instead allocation is through the market. Water rights are divided between permanent and contingent rights and are governed by private civil law as private property (Grafton et al., 2011, p. 12; Bauer, 2005, pp. 151–2). Rights are then further divided by type of use; consumptive rights that include agriculture and human consumption and non-consumptive rights such as hydropower and aquaculture (Rayl, 2016, pp. 34–35). Critically, there is no explicit beneficial use clause, meaning that resolving conflicts between users can be difficult and it is hard to reclaim water for the environment (Grafton et al., 2011, pp. 7–8). Due to a lack of infrastructure, difficulty in managing trades across regions (again the issues of abstracting water from its hydro-social conditions), problems with registration of water licences, and a local reluctance to participate, trading has been limited to certain geographical spaces such as the Limari Valley where water is more scarce and therefore more economically valuable (Grafton et al., 2011, p. 11).

4. The articulation of WTMs to circuits of global finance: Emerging tendencies and potential barriers

Despite some initial excitement among financial actors, the water futures market has been slow to take off. As Willem Buiter suggested at the start of this article (Buiter, 2021), the futures market may have been 'somewhat premature' because what is traded is not 'a single, homogeneous commodity or asset.' In other words, it may be its own fundamentals that are limiting expansion; the amount and quality of water is hard to define and predict meaning the contract (and what is to be traded) remains ill-defined. Moreover, at this stage a global water futures market based on physical settlement seems unlikely as water is highly localised and costly to transport[11]; or put differently, the material reality of the hydro-social cycle frustrates the process of re-engineering water as an abstract financial asset. However, futures markets based on cash settlement (such as this one) do not necessarily face the same obstacles, as what is being traded is the promise of an amount of water in the future that can be settled with cash rather than the resource itself; the promise of future water is rendered as an abstract financial value. Yet these barriers are neither new nor unique to the futures market as they can also be found in the fundamentals of each of the more established and longstanding WTMs explored in the previous section. Moreover, barriers are not necessarily limits to expansion (see Lebowitz, 2003), but may rather point to the next potential "problems" that can be overcome, through for example new rounds of financial re-engineering. It is these tendencies and contradictions that I will now turn to.

4.1. An increasing role of the state

In each WTM there is an extensive and increasing role of the state and related institutions as buyer and seller as well as creator and regulator of the market. While institutional regimes (such as legal systems and governance frameworks) must exist or be created so that water rights can be registered, respected, and managed (Meran et al., 2021, p. 186), the difficulty of determining what is being traded, by whom and to where, has necessitated increasing government regulation and oversight to function. For example, state and Federal governments play multiple roles in the Australian market. They are regulators, creating the water policies and legislation that the market operates through, they determine the amount of water that can be allocated each year, and they are also market actors buying back water entitlements (or through forms of reverse auction) on behalf of the environment. There is a tense but entangled relationship between the often-private exchanges where water is traded, and the layers of governmental regulation and bodies that determine what exactly is traded and how.

This reflects arguments about the breakdown of the public–private dichotomy of water services and infrastructures due to financialization that others have pointed to (see Bresnihan, 2016, p. 116; Furlong, 2020, p. 574). As Gibson has argued (Gibson, 2021, p. 98) water is being financialized through the articulation of financialized institutional logics to environmental governance, mediated by state institutions. In these WTMs it is through state rationales, institutions, and regulations that water is being reengineered as a potential financial asset in often contradictory and incomplete ways. Where water justice movements

[11] Although in the same article, Buiter refers to Project Greenland a project that aims to transport Greenland's glacier melt and icebergs to water scarce states in the MENA region (Buiter, 2021; Project Greenland, 2023).

have successfully politicised privatisation and many forms of public–private partnership (see Bieler, 2014; Muehlebach, 2023), the current murkiness of who is responsible, who profits and how, is much harder to follow and subsequently fight against.

4.2. Measurement, quantification, and standardisation

The increasing need for state institutions and oversight is linked to the problems of definition: Both what is being traded in each WTM and the tools and logics used to measure and monitor water. As Muehlebach suggests (Muehlebach, 2023, pp. 31–32) in reference to the financialization of water services 'All sorts of unlike entities—water, pipes, managerial systems, and nature—are reworked to appear as quantified, standardized, and priced, rendering their value commensurate and thus comparable and movable across time and space…'. Reflecting the depoliticised, technocratic and often fiercely resisted water governance mechanisms that characterise neoliberal water management (see Budds, 2009), these attempts to standardise what water is (i.e., what can be exchanged) (see Linton, 2010), alongside the unbundling of land and water property rights so that water licences can become an exchangeable commodity, are also necessary pre-conditions for financialization by abstracting water from the particularities of its hydro-social cycle.

However, this process of abstraction is often contested because in the search for an exchangeable form the socio-political values and rationales that water is mediated through are necessarily ignored. The financialization of nature, like coal mining or industrial farming that came before it, requires seeing nature in a new way. For example, the unbundling or segmentation of related "natures" such as land or water, or the abstraction of water from its hydro-social conditions into separate profit-making streams i.e., eco-system services in the form of aquifer recharges, water pollution trading, WTMs or futures trading can all occur through the same body of water, necessarily obscures the relations between those sources and processes. This can be a violent process which also creates new frontiers to be expropriated. For example, standardised water licences erase Indigenous understandings of water that run counter to segmentation and dualisms. It is, as we can see in the inability for the Australian WTMs to account for First Nations' cosmologies, a violence of erasure, or as Tati Tati Latji Latji, Wadi Wadi, Mutti Mutti man Brendan Kennedy argues a form of aqua Nullius (Kennedy et al., 2022); Indigenous value systems were not, and cannot, fully be accounted for within the given terms of the debate.[12] To access Cultural Flows that will allow for water to flood billabongs and floodplains necessary for the protection of country, the Tati Tati People of the North-West Murray River Country must navigate multiple levels of bureaucracy and what was once a natural process is now controlled by a water regulator (O'Donnell & Kennedy, 2023). Water flows that are dictated by regulators and limited to specific times, run counter to Tati Tati understandings of water as an 'ecological entity, a being and a sprit… water is essential to creation and Dreaming, with many ancestral beings created by and dwelling within water.' (Tati Tati, 2022).

The process of standardisation and abstraction also sharpens other contradictions inherent to the application of economic abstractions and financial valuations to nature. Within each of the explored WTMs – and seen through capitalist logics in general – water maintains two contradictory values, it is both a dead and static resource that can be expropriated for irrigation, fracking or similar, but is also an eco-system service with its value determined by its capacity to regenerate. Yet these are two seemingly mutually exclusive valuations. Building on Bracking's argument (Bracking, 2020, p. 216) the animism of water's materiality and its necessary interconnectivity with that which exists around and through it (land, infrastructures, the broader hydro-social cycle) frustrates economic calculations. While from the vantage point of those creating new water-based financial products, the financialization of water is a relatively smooth process with water futures markets one logical conclusion, from another vantage point, water's animism is disassembling financialization. Financial valuation requires nature to be quantifiable, understandable, and exchangeable; it takes a single snapshot of a necessarily living organism, which in freeze-framing also seeks to erase those connections and animate qualities that cannot be readily accounted for within existing (especially economic) models, even though it is these animate qualities that give water its use value. Financialization is, as Bracking argues (Bracking, 2020, p. 216), an anti-ecological position. These underlying logical contradictions inherent to the process of abstraction necessary for financial valuation cannot be papered over.

4.3. Narratives of environmental protection

Processes of separation, measurement and monitoring are not only required to make water legible to financial markets but are also necessary to address over-use and environmental degradation. What we see in each WTM is an increasing articulation of arguments of environmental protection through mechanisms of financialization such as speculation, marketisation, and securitisation. Water licences have evolved into a form of speculation through environmental risk management, as what is traded is access to a future (somewhat undetermined) given pool of water for a set amount of time. Another common tension is whether the environment is one "equal" actor of many in the market, or if it has a special status and if so how to manage market equivalence. Rising political tensions and push-back when environmental buy-backs have been proposed in Australia are but one example. Further highlighting the entanglement of measurement, environment and the state, this complexity often requires further regulatory expansion through state institutions and new policies (especially in Australia).

WTMs have also been linked with social and environmental impact investing. For example, groups such as Nature Conservancy have developed financial products that link environmental and social impact to economic returns through the Australian WTM; Water Sharing Investment Partnerships, first launched in Australia in 2015, invested $AUD27 million in environmental impact investments in 2016 (Richter, 2016, p. 16). With the growth in green bonds and discussions on green finance at the recent UN Water Conference, there may be further financial incentives to link existing water infrastructures such as a treatment plant, pipelines, or WTM to the green economy if green credentials can be established.

This is despite the case that the environmental impact of WTMs is contested.[13] It is questionable whether profit-seeking market-based models can be mobilised for environmental protection in the long term as increasing investment due to rising prices relies upon prices and revenue streams (in WTMs the price of water licences) continuing to increase. Yet such price increases are to some degree predicated on increasing scarcity. This highlights a contradiction in the underlying logic of utilising WTMs to improve environmental standards; the economic models are designed to encourage water savings, yet at the same time rely on increasing scarcity for those same actors to make a profit. This tension is also reflected in the problems regulators have faced when encouraging water savings in commercialised water services,[14] as increased savings would decrease revenue for commercialised water companies when water bills are determined through consumption rates

[12] There are a number of interesting research projects that have tried to work out how Cultural Flows, for example, could be embedded within the Murray Darling Basin Plan and the Australian WTM (see Australian Cultural Heritage Management, 2014; L. Hartwig & Jackson, 2020; Kennedy et al., 2022).

[13] Although hard to distinguish the impact of water trading from the broader Murry Darling Basin Plan, the Australian State of the Environment Report in 2017 reported that inland water was very poor and deteriorating (Grafton, 2019, p. 123).

[14] English water companies are a key example.

i.e., metering. There is a built-in disincentive for water companies to encourage water savings at a time when water savings (through leaks, changed behaviour…) are most required. As such, there is a clear problem if water governance is reliant upon mechanisms designed also for profit-maximisation where potential profit is predicated upon continued scarcity, whether through water licences, water bills or the securitised versions of them.

4.4. "Equity" in the market

At each point in the hydro-social cycle – from water services, infrastructures, lakes or aquifers – neoliberal and subsequent financialised water governance has tended to result in rising water prices and shifting waterscapes that have benefited larger fractions of capital (Barlow, 2019).[15] This is also true for the cases of Chile, Australia and western USA where the waterscape has been transformed to the benefit of larger irrigators and producers and fuelled water-based conflicts. In Chile, the creation of non-consumptive water rights has encouraged hydroelectric power development and there are increasing conflicts over speculation and the resultant concentration of wealth (Borgias, 2018). Irrigation efficiency has also remained low in Chile despite market incentives and there is growing critique of the inability for the current design to address ecological flows (Rayl, 2016, pp. 40–41). There have been cases in the US where the buying up of water licences by financial actors – including state-backed investment funds and municipalities – has dramatically altered what is grown in these areas and has fuelled conflicts between water users as the draining of groundwater resources has rendered some areas uninhabitable (Halverson, 2023).

In Australia, there is growing critique of the way that irrigators and speculators have shaped the water market. As water trading in Australia occurs on private exchanges there is little public record of who is buying and selling entitlements and to what extent speculators have impacted prices. However, some initial research suggests there is increasing foreign ownership of water rights (Schremmer & Isa, 2019), and what is clear is that large amounts of capital have flowed through the market; it is estimated that between 1995 and 2008 the price of water licences in the MDB increased annually by 15 % (O'Brien, 2008, p. 527) and about $AU6billion of trades were made in the 2020–21 financial year (Bureau of Metereology, 2022, p. 4). Financial benefits have primarily flowed to large irrigators backed by transnational capital who thus have better liquidity than smaller farmers. As Grafton asserts (Grafton, 2019, pp. 130–131), there has been a degree of regulatory capture of water policy by large irrigators and lobbyists especially in the Northern Basin. This has facilitated the consolidation of agricultural production, with one report finding that four irrigators were responsible for 75 % of the water extracted in the upper MDB (Davies, 2019a). Banks have invested in water and new water infrastructure, and have offered loans backed by water rights (Dingle, 2018). Buy-backs for the environment have also been muddied, with the now infamous case of $AU80 million paid to a company connected to the then Liberal Party Energy Minister, Angus Taylor, for low quality overland flow entitlements that are yet to yield any water (Davies, 2019b). Murray Darling Basin Traditional Owners have also critiqued WTMs arguing that water trading is reliant on an understanding of water that is in opposition to their understanding and relationship to water (Australian Cultural Heritage Management, 2014), and that as Aboriginal water holdings operate outside the market they are also excluded from any potential economic benefits (Hartwig et al., 2018, p. 10).

Although the market is frequently touted as an equitable means through which water can be allocated it seems to have, whether intentionally or not, resulted in increasing water inequity, with access determined by who can pay. As the Special Rapporteur for the Right to Water and Sanitation, Pedro Arrojo Agudo states (Arrojo Agudo, 2021), it is not only water scarcity but also unaffordability that underpins water access issues. The argument that environmental, social, and economic interests can be equally addressed through economic models based on logics of profit-maximisation are inherently contradictory: Where increased water prices may encourage water savings and more efficient water use for those that can afford to shift production models or invest in water saving technologies, such price increases will restrict water access for those that cannot. As such, when models are designed for profit-maximisation rather than equity it is likely that the economic logics that underpin them, rather than the social and ecological issues that they seek to address, will win out.

5. Conclusion: Financial flows through the hydro-social cycle?

As Ballestero reminds us (Ballestero, 2019, p. 414), water can be infrastructure, environmental asset or service, or an input to other production processes; water is entangled, multiple and material all characteristics that can frustrate its reduction to a financialised form. In the Chilean, Australian and US WTMs that were the focus of this study, water was first unbundled from land rights – abstracted from its particular hydro-social cycle – and through monitoring and regulations overseen by state or state-adjacent institutions water has been, to varying degrees of success, re-embedded in the hydro-social cycle as a standardised and marketized product that is increasingly legible to circuits of global finance capital. This process is often facilitated by narratives of environmental protection, efficiency, and equity of access. The water futures market should be approached as an intensification of these processes rather than a novel moment in the evolution of water governance. It is a clear example of water being re-engineered as an asset class and embedded within global financial markets, even if, as Buiter suggests, it continues to face material barriers to its expansion due to the specific qualities of water and the hydro-social cycle.

Ultimately, current water governance trends rest on an assumption that environmental, social, and economic values are compatible. From this, the policy solution is defined as creating enabling conditions that would encourage private financing. This has included institutional re-engineering that prioritises investment regulations, transparency, and the strengthening of protections for private property. It has also included the financial re-engineering of infrastructures and debt obligations as Pryke and Allen explore (2019), alongside the mobilisation of social or green impact investing (Stolz & Lai, 2020). Each of these processes seek to quantify and standardise dynamic water resources in ways that either erase the particularities of water (which make exchange difficult) or accommodate for those differences through logics of risk management and ratings. While these logics are necessary for exchange, they also necessarily erase the specific ecological, social, and political dimensions that underpin the global water crisis that such regimes are nominally seeking to address.

Drawing out, this suggests that an increasing financialization of global water governance – the potential emergence of a financialised hydro-social regime – is murkier in comparison to the neoliberal hydro-social regime that came before it. Building on the arguments of Bresnihan and Acevedo Guerrero et al. (Acevedo Guerrero et al., 2015; Bresnihan, 2016, 2020) the public–private dichotomy of who manages and owns water is harder to maintain than it was during periods of outright privatisation. The state is a site of financial investment, regulator, and increasingly also a financial investor. And, although profit has been a driving force through each period of water governance, this is now being increasingly articulated through narratives of environmental protection and equity of access (especially the SDGs).

In bringing these tendencies together and looking towards what might come next, what I want to suggest is that processes of financialization are concurrently gaining a foothold at multiple nodes across the hydro-social cycle in ways that could facilitate the financialization of other nodes. For example, if I were to start from a different node in the

[15] The infamous example of Macquarie Group and Thames Water exemplifies this (Barrett, 2023).

hydro-social cycle, such as water infrastructures or services we can also find ready examples of how they are being financially re-engineered (see Barrett, 2023; Pryke & Allen, 2019). A muddying of public–private actors and investment, a mobilisation of environmental protection narratives, and a language of risk management and speculation are also being employed.

If we understand the hydro-social cycle as a *cycle* what comes into view is not just the way that water flows from aquifer to pipeline, to treatment plant and then industry, but also the way that financial logics and potentially profit are also flowing through the same channels, facilitating the further financialization of each node while doing so. For example, increasing water licence prices due to speculation in WTMs may increase the overall price of water in specific areas, and with increasing water prices, desalination plants may become financially viable where they previously were not as the costs of desalinated water has tended to be much higher than non-desalinated sources.[16] With an equalising of water prices across multiple sources, there could be a flurry of investments in new water infrastructures from desalination plants to transportation pipelines. These dynamics are already evident in the US, where increasing prices for water (or specifically water licences) have made proposed pipelines newly profitable as when water prices were lower the cost to build the pipelines outweighed any potential profits from the water that would have been transported through them.[17] While I am not arguing that these new desalination plants or pipelines are solely due to increasing water licence costs in WTMs, there is clearly a interdependence that needs to be further explored, and previous research has tended to take each example of financialization as a standalone case rather than an interdependent node in a wider financialized hydro-social regime.

As the intensification of financialization increases we could see some of the material barriers that Buiter points to be resolved. The futures market could pave the way for further consolidation of formerly separate WTMs rectifying previous barriers of locality. Or learning from Nature Conservancy in Australia, claims that environmental conditions can be improved through water trading could be further re-packaged as an eco-system service and subsequent green bond. Ultimately, with water prices rising due to the ecological crisis of which the global water crisis is one facet, potential revenue streams for these new infrastructures may improve, while also allowing for these infrastructures to be integrated into green bond markets, climate bonds, or social impact investing. While diverse, each process of financialization tries to smooth out differences between places, scales, and nodes across the hydro-social cycle. Therefore, although the material limits that Willem Buiter points to when he states that there is not yet a standard global water product remain i.e., we may never overcome the materiality of water, this does not discount the capacity for financial actors and processes to continue to attempt to smooth over and abstract from these conditions to increasing degrees.

As others have argued (Swyngedouw, 2009, p. 57), hydraulic environments are socio-physical constructions and are therefore, not politically, socially or ecologically neutral. This article has demonstrated how financial logics are shaping global water governance in particular WTMs in unequal ways, and at the same time have come up against the materiality of water through doing so. While this article started from WTMs in order to point to these trends, future research is needed to systematically explore how these processes may feed into, facilitate, and potentially hinder one another across the hydro-social cycle. As it may be the way that these processes flow between each node that become critical to the future of, and resistance to, water financialization.

[16] The Claude "Bud" Lewis Carlsbad Desalination Plant in San Diego County is one of the largest and technologically developed desalination plants and operates as a public–private-partnership providing water to San Diego.

[17] The Vista Ridge Pipeline Project in San Antonio, Texas is one example.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements:

The author would like to acknowledge the many discussions she has had with water activists, and water researchers, alongside her work with the Special Rapporteur for the right to water and sanitation, that helped shape these arguments. Earlier versions of this paper received valuable feedback at the CPERN conference, AIPEN, and a panel at the ISA that formed the basis of this special issue. Special thanks go to Kate Bayliss, Philip Mader, Pranjal Deekshit, and Michael Pryke for valuable feedback and organisation of this special issue, alongside the constructive feedback from two anonymous reviewers. Thanks also to Max whose capacity to be a good napper allowed her to finish the necessary revisions while on maternity leave. A grant from the Department of Sociology, Bielefeld University for early career researchers was used to carry out the initial research for this article.

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